

Question_10:

Code:

```
1  /*
2  Question_10: Find the number of distinct values present in subarray A[L...R].
3  */
4
5  #include <iostream>
6  #include <vector>
7  #include <cmath>
8  using namespace std;
9
10 long long freq(long long x) {
11     return x * (long long)sqrt(x) + (x + 1) / 2;
12 }
13
14 int main() {
15     int Q;
16     cin >> Q;
17
18     vector<long long> start(1, 1), finish(1, 0);
19     long long pos = 1;
20
21     for (long long x = 1; pos <= 10000000000000LL; x++) {
22         long long f = freq(x);
23         start.push_back(pos);
24         finish.push_back(pos + f - 1);
25         pos += f;
26     }
27     while (Q--) {
28         long long L, R;
29         cin >> L >> R;
30
31         int lo = 1, hi = (int)start.size() - 1;
32         while (lo < hi) {
33             int mid = (lo + hi) / 2;
34
35             if (finish[mid] >= L)
36                 hi = mid;
37             else
38                 lo = mid + 1;
```

```
39     }  
40     int first = lo;  
41  
42     lo = 1;  
43     hi = (int)start.size() - 1;  
44  
45     while (lo < hi) {  
46         int mid = (lo + hi + 1) / 2;  
47         if (start[mid] <= R)  
48             lo = mid;  
49         else  
50             hi = mid - 1;  
51     }  
52     int last = lo;  
53     cout << max(0, last - first + 1) << '\n';  
54 }  
55 return 0;  
56 }
```

Output:

Test Case_1:

```
2  
3 3  
1  
2 6  
3  
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
3  
3 3  
1  
2 6  
3  
5 4  
1  
PS C:\Users\Prabin Yadav\
```